Data Selection and Full-Parameter Supervised Fine-Tuning for Qwen2.5-3B Reasoning

Rohit Kumar Salla Manoj Saravanan Haissam Adam
Reasoning SFT Course Project

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Abstract

We study a completed full-parameter supervised fine-tuning run of Qwen2.5-3B-Instruct using 15,000 examples selected from the assignment's AceReason corpus. The selection procedure combines answer-completeness and repetition filters with topic clustering and lexical reasoning-route coverage. We compare the resulting model with the original base on nine reasoning benchmarks using identical sampling and grading settings: eight responses per question, temperature 0.6, top-p 0.95, and up to 32,768 generated tokens within the configured context window. On the assignment's seven-benchmark average, pass@8 increases from 49.55% to 51.52%, while pass@1 decreases from 23.55% to 21.85%. The recorded paired confidence interval for the pass@8 difference includes zero. Truncated samples increase from 141 to 1,964 among 15,792 responses per model. These findings indicate that the apparent average gain depends on the evaluation metric and accompanies substantially worse stopping behavior. We document the selection algorithm, actual 4,096-token training context, training dynamics, per-benchmark uncertainty, and limitations of attributing the result to data selection without a matched random-data training control.

1 Introduction

Supervised fine-tuning provides a direct way to adapt an instruction-following language model to mathematical and scientific reasoning demonstrations. In a fixed training-data budget, the selection of examples is an important experimental choice: selecting many near-duplicate solutions may offer little topic coverage, whereas prioritizing only difficult or long demonstrations may create a mismatch between the available training context and the responses being taught. The present project examines a practical selection pipeline and reports the behavior of its completed model rather than assuming that longer solutions or lower training loss imply better reasoning.

The starting model is Qwen2.5-3B-Instruct, an instruction-tuned causal language model with approximately 3.09 billion parameters [1]. Training uses LLaMA-Factory [3] and a 15,000-example subset derived from the assignment's AceReason-100K source [2, 4]. The base and fine-tuned models are evaluated on identical questions and under matching generation settings. This comparison measures the combined effect of the selected examples and the stated training configuration; it does not separately identify the causal contribution of each selection component.

The report addresses three questions. First, how can source-text features be used to obtain a topic-balanced and stylistically varied subset? Second, how does the completed model perform on each benchmark and on the assignment's specified average? Third, does performance under multiple sampling agree with single-sample accuracy and response termination? These questions motivate reporting the full score table, paired intervals, training traces, and truncation behavior together.

2 Data and Selection Method

2.1 Corpus Preparation

The input is the supplied AceReason-100K dataset [2]. Records are normalized into an instruction–response representation with optional input and system fields. The preparation code removes invalid records, deduplicates normalized prompts, and excludes matching benchmark questions. Near-overlap filtering uses normalized character shingles with a minimum-overlap criterion. Validation examples are reserved before selection, and near-duplicates of validation questions are removed from the candidate pool.

The resulting random and advanced exports each contain 15,000 examples. The random export samples from the prepared candidate pool with seed 42. The advanced export applies additional eligibility and coverage criteria. Consequently, the two exports differ in both filtering and sampling; they should not be treated as differing solely in cluster allocation.

2.2 Eligibility Filters

Token counts are measured using the Qwen chat template. An advanced-selection example must have at least 128 response tokens, a detected complete boxed answer, and a repeated-word-eight-gram fraction no larger than 0.35. The full serialized sequence must fit the 8,192-token selection limit with a 32-token reserve. The repeated-eight-gram fraction is

$$r(y) = 1 - \frac{|\operatorname{unique}(G_8(y))|}{|G_8(y)|}, \tag{1}$$

where $G_8(y)$ is the sequence of contiguous word eight-grams; the value is defined as zero when no such eight-grams exist. These filters measure observable properties of demonstrations. A boxed answer and low repetition do not certify mathematical correctness.

2.3 Topic Coverage and Route Features

Topic features are obtained from prompt text. The default implementation uses character TF–IDF features, truncated singular-value decomposition, and MiniBatch K-means clustering. For the selected advanced export, the quota table contains 113 populated eligible clusters. A greedy square-root allocation balances corpus prevalence and broader coverage. If cluster j has capacity N_j and current quota q_j, each additional slot is assigned to the non-full cluster maximizing

$$\frac{\sqrt{N_j}}{q_j + 1}, \qquad \sum_j q_j = 15{,}000. \tag{2}$$

This allocation reduces the dominance of very large clusters while respecting each cluster's available examples.

Within each cluster, coverage is measured using a 180-dimensional lexical route vector. Twelve cue families cover checking, revision, alternative approaches, case reasoning, conclusions, modular arithmetic, geometry, counting and probability, algebra, calculus, invariants and proof techniques, and enumeration. The vector concatenates normalized cue counts, the fraction of each cue appearing in the latter half of the response, normalized cue-to-cue transition counts, and cue-presence indicators. This representation describes solution text rather than an inferred proof graph.

The vectors are standardized over the eligible corpus, clipped to the range $[-5, 5]$, and normalized before a greedy farthest-point procedure. The first example maximizes a low-repetition quality score. Subsequent choices combine distance to the nearest selected example with quality:

$$U_i = 0.8 \min\left(\frac{d_i}{2}, 1\right) + 0.2(1 - r(y_i)), \tag{3}$$

where d_i is the nearest squared distance in normalized route-feature space. Selected examples are shuffled with seed 42. The saved response-token totals are approximately 62.87 million for the advanced export and 109.50 million for the random export. These are export-level counts, not counts of tokens actually supervised by the trainer.

3 Training Setup

3.1 Objective and Configuration

Training updates all model parameters using the response-only autoregressive cross-entropy objective. For input x, target response y, and the set $\mathcal{S}$ of supervised response positions retained after preprocessing,

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} \log p_\theta(y_t \mid x, y_{<t}). \tag{4}$$

Prompt positions are excluded from the loss. The run uses the Qwen template, BF16 mixed precision, SDPA attention and gradient checkpointing. Packing is disabled. Table 1 lists the actual configuration of the submitted model.

Table 1: Training configuration of the completed model.

Setting	Value	Setting	Value
Training examples	15,000	Context length	4,096
Optimizer updates	1,875	Batch per device	1
Gradient accumulation	8	Effective batch	8
Learning rate	2×10^{-6}	Schedule	Cosine
Warmup ratio	0.03	Weight decay	0.01
Gradient clipping	1.0	Seed / data seed	42 / 42
Training hardware	One B200	Precision	BF16 mixed

The trainer completed 1,875 optimizer updates and reported epoch 1.0. With an effective batch size of eight, this corresponds to one traversal of 15,000 examples. The recorded runtime is 3,989.39 seconds, or 66.49 minutes; this is the trainer-reported runtime and excludes separate preprocessing, scheduling and final benchmark evaluation. The reported mean training loss is 0.5780. Recorded loss and gradient-norm values are finite throughout the completed history.

3.2 Context and Loader Considerations

The 8,192-token data-selection cutoff differs from the actual 4,096-token training context. Eligible demonstrations can therefore pass selection but be truncated during training, potentially removing part of the final derivation or answer. The exported dataset should not be described as 8,192 tokens of supervised context per example. The original dataset configuration also does not explicitly map the system column. The report retains these details as limitations of this completed run; it does not infer a measured causal effect from either mismatch.

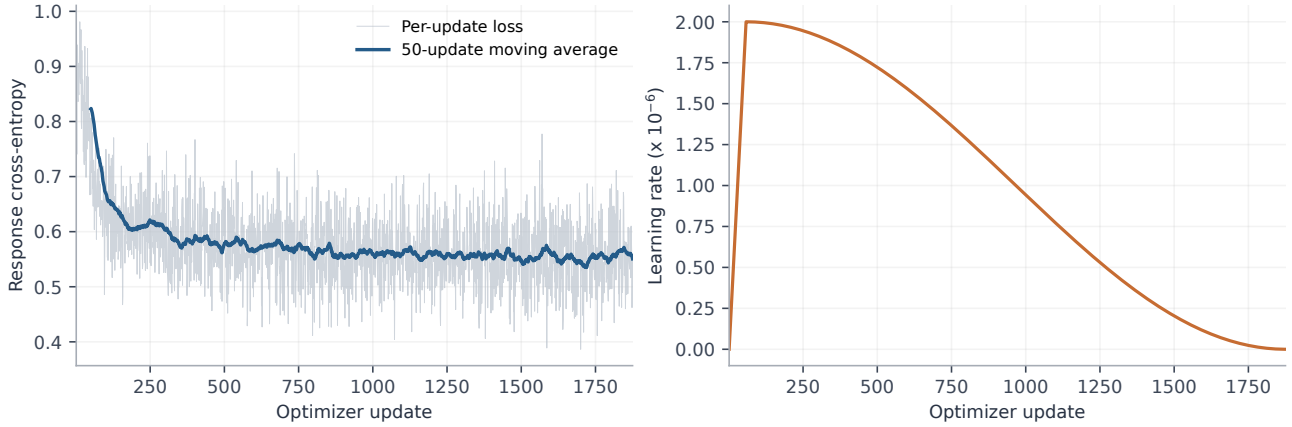


Figure 1: Recorded training dynamics for all 1,875 optimizer updates. The loss panel shows raw per-update response cross-entropy and a trailing 50-update moving average. The learning-rate panel shows the actual warmup and cosine schedule. These traces establish optimization progress; they are not estimates of benchmark accuracy.

4 Evaluation Protocol

4.1 Benchmarks and Generation

The benchmark suite contains AIME 2024, AIME 2025, MATH-500, CN Math 2024, Kaoyan, AMC, Minerva Math, OlympiadBench and GPQA-Diamond. The recorded suite contains 1,974 records. Each model generates eight answers per record, yielding 15,792 sampled responses per model. AIME 2025 was separately sourced as a supplement to the supplied suite. The MATH evaluation here contains 500 records; reported denominators use the actual evaluated records rather than the count printed in the assignment's introductory table.

Both arms use temperature 0.6, top-p 0.95, at most 32,768 new tokens, a configured maximum model length of 32,768, seed 0 with stable question-specific seeds, batch size 1,024, at most 128 active sequences, tensor parallelism one and GPU-memory utilization 0.85 on one B200. The system instruction requests step-by-step reasoning and a boxed final answer. Answer extraction and rule-based grading follow the assignment implementation [4]. The model context includes the prompt, so the number of available output tokens can be smaller than the new-token limit.

4.2 Metrics and Aggregation

Let c_i be the number of correct sampled responses among $n = 8$ for question i. The recorded pass@1 is the average sample correctness, while pass@8 is the fraction of records for which at least one sample is correct:

$$\text{pass@1} = \frac{1}{N}\sum_{i=1}^{N}\frac{c_i}{8}, \qquad \text{pass@8} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}[c_i > 0]. \tag{5}$$

The assignment's AVG is an unweighted mean over seven benchmarks: AIME 2025, CN Math 2024, Kaoyan, AMC, Minerva Math, OlympiadBench and GPQA. AIME 2024 and MATH are reported individually but are not part of that average. This mean is distinct from pooling all questions across datasets or averaging all nine benchmarks.

4.3 Paired Uncertainty

Saved per-benchmark intervals use a paired bootstrap with 2,000 resamples and seed 42. Questions, rather than individual sampled answers, form the resampling units. Repeated normalized question records are clustered together while preserving their record weights. The same resampled units are used for both models. The intervals are descriptive, per-comparison intervals without a multiple-comparison correction, and they do not measure variation across training seeds. Table 3 lists the recorded per-benchmark intervals.

5 Benchmark Results

Table 2: Complete benchmark results in percent. Each score is recomputed from the saved eight-sample correctness vectors.

		Pass@1		Pass@8	
Benchmark	Questions	Base	Fine-tuned	Base	Fine-tuned
AIME 2024	30	7.08	4.58	23.33	16.67
AIME 2025	30	2.50	3.33	13.33	20.00
MATH-500	500	65.88	64.00	85.20	85.00
CN Math 2024	30	7.50	7.92	23.33	33.33
Kaoyan	199	22.42	21.23	56.78	59.80
AMC	40	44.38	37.50	77.50	70.00
Minerva Math	272	31.07	27.30	49.63	48.90
OlympiadBench	675	27.50	26.33	49.48	49.33
GPQA-Diamond	198	29.48	29.36	76.77	79.29
Seven-benchmark mean	—	23.55	21.85	49.55	51.52

The seven-benchmark pass@8 mean increases by 1.97 percentage points, from 49.55% to 51.52%. The largest positive numerical differences are on CN Math 2024 and AIME 2025, both of which contain only 30 questions. Their pass@8 differences of 10.00 and 6.67 percentage points correspond to three and two additional solved records, respectively. These small denominators make the displayed percentages sensitive to a small number of outcomes.

Kaoyan and GPQA also show higher pass@8 after fine-tuning. Conversely, AIME 2024 and AMC decrease, while MATH, Minerva and OlympiadBench change slightly downward. The observed result is therefore not a uniform performance gain. The assignment mean reflects the weighting rule applied to these mixed changes rather than a general improvement on every reasoning domain.

The recorded paired interval for the seven-benchmark pass@8 difference is $[-1.89, +5.74]$ percentage points. Because it contains zero, the experiment does not establish a statistically conclusive average pass@8 gain. The corresponding recorded pass@1 interval is $[-3.16, -0.22]$ points for a mean difference of -1.70 points. These intervals describe uncertainty conditional on the completed runs and the evaluation samples, not the uncertainty of retraining the entire method.

6 Generation Behavior and Interpretation

The multiple-sample metric and single-sample metric move in different directions. Average pass@1 decreases from 23.55% to 21.85%, even though average pass@8 increases. A model can become more likely to produce at least one correct result across several attempts without increasing the expected

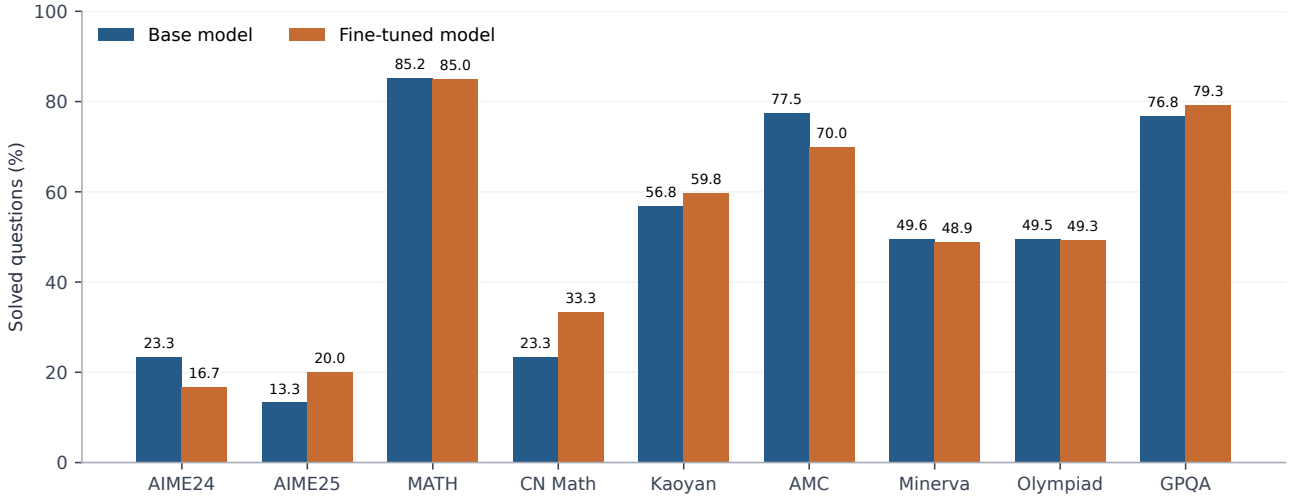


Figure 2: Base and fine-tuned pass@8 on all evaluated benchmarks. Each bar represents the fraction of question records solved by at least one of eight generated answers. The full table includes both improvements and regressions.

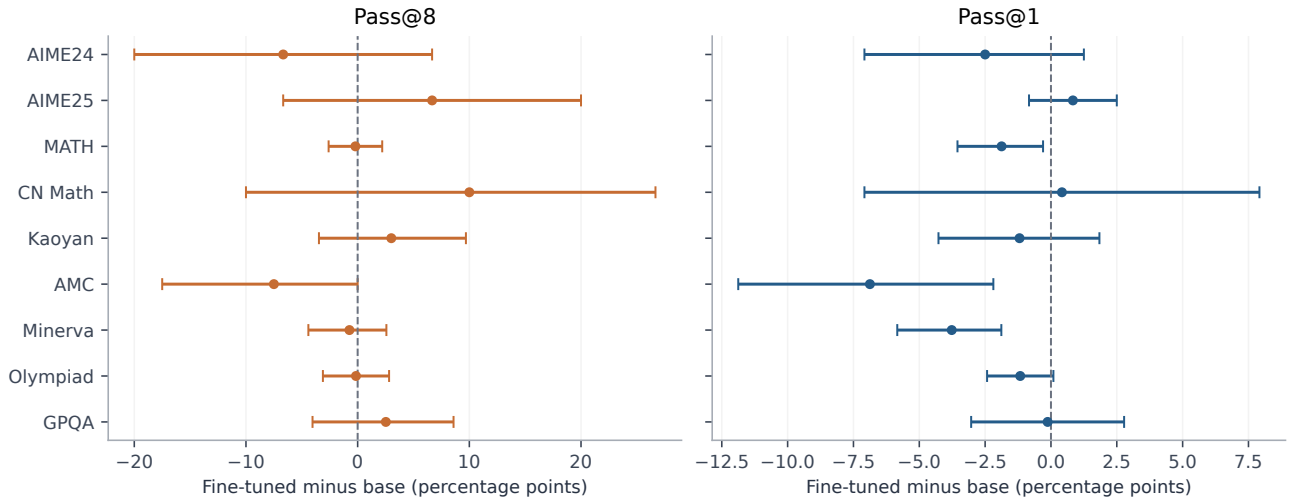


Figure 3: Recorded paired 95% bootstrap intervals for benchmark-level differences. The dashed line denotes equal scores. Intervals resample question clusters and are not adjusted for multiple comparisons. The plotted quantities are percentage-point differences, not relative percent changes.

correctness of a randomly chosen attempt. This experiment records that tradeoff, but it does not identify its mechanism: diversity of reasoning paths, answer variability, token-budget use and sampling effects were not separately manipulated.

Termination changes substantially. The base model has 141 truncated responses across the full suite, compared with 1,964 for the fine-tuned model: approximately 0.89% and 12.44% of the 15,792 sampled responses. Truncation measures responses recorded as reaching a length boundary. It does not by itself establish whether those responses would eventually become correct, nor does it account for all computational cost.

The sharp increase in truncation is relevant to usability: a higher multiple-attempt solve rate does not guarantee efficient or dependable individual answers. Training on longer source responses while restricting the training context to 4,096 tokens is a plausible concern, but the present evidence does not

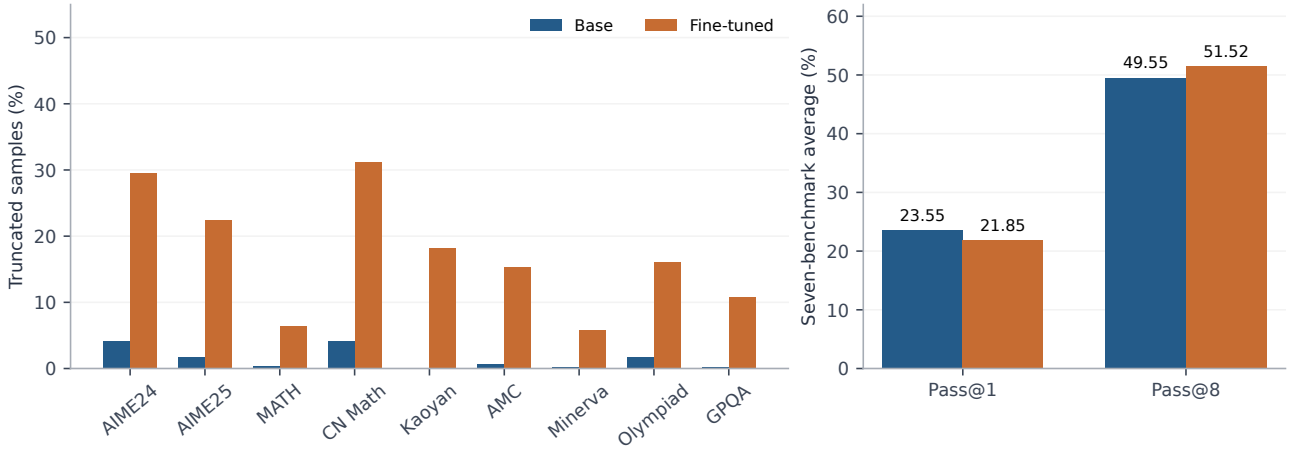


Figure 4: Generation behavior and the metric tradeoff. Left: truncated samples as a percentage of sampled responses within each benchmark. Right: the specified seven-benchmark mean for pass@1 and pass@8. The panels describe the same completed evaluation rather than a separate experiment.

isolate that factor. It would be inappropriate to attribute the observed termination behavior solely to truncation, the loader mapping, the selection rule or a particular hyperparameter without a controlled comparison.

7 Limitations and Reproducibility

No matched random-data training control. Both random and advanced datasets are distributed, but a completed full-parameter random-data model evaluated under this same nine-benchmark protocol is not available. The base-versus-SFT comparison measures a complete training pipeline, not the advantage of advanced selection over random selection. A fair selection ablation would hold initialization, training budget, context, optimizer, sampling and grading fixed while varying the selected data.

Single training seed and adaptive exploration. The report concerns one completed training seed. Benchmark resampling does not substitute for multiple training seeds. The model comes from a broader project with other experiments, so the present evaluation should not be treated as a preregistered confirmation study.

Proxy-based selection. Route vectors detect lexical patterns, not the validity or causal structure of reasoning. Quality filters favor complete-looking answers and low repetition but cannot guarantee correctness. Cluster quotas also depend on the chosen text representation and clustering procedure. The experiment does not establish that these particular design choices are optimal.

Data and metric boundaries. Overlap removal reduces recognizable contamination but does not prove absence of all semantic overlap or exclude pretraining exposure. AIME 2025 is separately sourced, and results should be interpreted using the actual evaluated records. Rule-based grading and boxed-answer extraction can also differ from human judgments of reasoning quality.

Artifacts. The public project repository includes the configuration, selection/evaluation code, recorded scores, per-question correctness vectors, this LaTeX source and plot-generation script. The model and both 15,000-example datasets are hosted on Hugging Face. The result-checking script recomputes all benchmark scores from the saved correctness vectors. The public download script uses a readable release tag. Original model, dataset and software license terms remain applicable.

8 Conclusion

The completed 15,000-example full-parameter SFT run produces a mixed reasoning result. The seven-benchmark pass@8 mean rises numerically to 51.52%, but its recorded confidence interval does not establish a conclusive gain over the base. Average pass@1 declines, and length-limited outputs become substantially more common. Reporting these outcomes together is essential: average multi-attempt accuracy alone would obscure a material change in individual-answer reliability and generation behavior. A matched random-data control, consistent selection/training context and repeated training seeds are the most direct next experiments for evaluating whether the proposed selection procedure offers a reproducible benefit.

References

[1] Qwen Team. *Qwen2.5-3B-Instruct: Model card*. Hugging Face, 2024. https://huggingface.co/Qwen/Qwen2.5-3B-Instruct.

[2] Assignment dataset maintainers. *AceReason-1.1 100K subset*. Hugging Face. https://huggingface.co/datasets/redsgnaoh/acereason11_100k.

[3] Yaowei Zheng et al. *LlamaFactory: Unified Efficient Fine-Tuning of 100+ Language Models*. arXiv:2403.13372, 2024. https://arxiv.org/abs/2403.13372.

[4] REDS Lab. *Project-Reasoning-SFT-LLM: Assignment, evaluator and submission instructions*. 2026. https://github.com/reds-lab/Project-Reasoning-SFT-LLM.

A Per-Benchmark Paired Intervals

Table 3: Recorded differences and 95% intervals in percentage points. Endpoints are from the saved paired question-cluster bootstrap comparisons.

	Pass@8		Pass@1	
Benchmark	Difference	95% interval	Difference	95% interval
AIME 2024	-6.67	[-20.00, 6.67]	-2.50	[-7.08, 1.25]
AIME 2025	+6.67	[-6.67, 20.00]	+0.83	[-0.83, 2.50]
MATH-500	-0.20	[-2.60, 2.20]	-1.88	[-3.55, -0.30]
CN Math 2024	+10.00	[-10.00, 26.67]	+0.42	[-7.08, 7.92]
Kaoyan	+3.02	[-3.47, 9.69]	-1.19	[-4.27, 1.84]
AMC	-7.50	[-17.50, 0.00]	-6.88	[-11.88, -2.19]
Minerva Math	-0.74	[-4.41, 2.57]	-3.77	[-5.84, -1.88]
OlympiadBench	-0.15	[-3.11, 2.81]	-1.17	[-2.43, 0.09]
GPQA-Diamond	+2.53	[-4.04, 8.59]	-0.13	[-3.03, 2.78]

B Software and Artifact Access

Training uses PyTorch 2.10.0, Transformers 4.55.0 and the vendored LLaMA-Factory implementation. Evaluation uses vLLM 0.19.0 and Transformers 4.57.6. Separate observed dependency lists are included in the repository; they are not complete dependency lockfiles. The training configuration uses no DeepSpeed configuration.